Stroke Prediction Using Random Forest: Dataset Analysis and Comparison with Published Results

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***Abstract*—This paper presents a Random Forest classifier for stroke prediction on the public Stroke Prediction Dataset (5,110 instances, 12 attributes), implemented in WEKA and benchmarked against published studies. After preprocessing and applying SMOTE to balance the minority class, the model achieved an accuracy of 96.60% using 10-fold cross validation and 95.6% using a 70/30 split. These results match or exceed the 95.70% and 94.11% reported in two of the reviewed studies under the same evaluation methods.**

Keywords—Random Forest, stroke prediction, SMOTE, WEKA, cross validation

I. Introduction

Stroke is one of the most prevalent and serious conditions in healthcare, and early identification and intervention are very important. Analyzing patient data can help in understanding warning signs and symptoms, which supports more effective treatment plans. The goal of this work was to build a Random Forest classifier in WEKA for the stroke dataset and benchmark its performance against published research. This paper presents my individual component of a group project on healthcare prediction classification.

II. Data

The public Stroke Prediction Dataset [4] was used. It is a structured, single-file CSV of clinical records containing 5,110 instances and 12 attributes, with a binary target attribute, Stroke (Yes or No). Example predictor attributes include ID, Gender, Marital status and Age. The machine learning task is classification.

III. Literature Review

I reviewed three papers on stroke prediction using imbalanced datasets. Singhai et al. [1] tested seven models on 5,110 records with 12 attributes, applying SMOTE to balance the 249 stroke cases; Gradient Boosting achieved the highest accuracy at 97.2%, while Random Forest reached 95.7%. Shobayo et al. [2] expanded their dataset to 9,722 instances using SMOTE and compared three models, with Random Forest performing best at 94.1%. Saputra et al. [3] also applied SMOTE to balance the same 5,110-instance dataset and compared six models, reporting Random Forest as the top performer at 82.1%. Together these studies show that while boosting methods can achieve the highest accuracy, Random Forest consistently delivers strong results across different stroke datasets and preprocessing strategies. Table I summarizes the studies.

Table I. Summary of stroke papers reviewed

Citation	Evaluation (split / CV)	ML algorithms	Selected	Top
Singhai et al. [1]	10-fold CV	LR, NB, DT, ETC, RF	RF 95.70%	GBC 97.2%
Shobayo et al. [2]	70/30 split	LR, DT, RF	RF 94.11%	RF 94.11%
Saputra et al. [3]	10-fold CV	LR, DT, RF, KNN, NB, SVM	RF 82.07%	RF 82.07%

RF: Random Forest, LR: Logistic Regression, NB: Naïve Bayes, DT: Decision Tree, ETC: Extra Trees Classifier, GBC: Gradient Boosting Classifier, KNN: K-Nearest Neighbors, SVM: Support Vector Machine.

IV. Analysis and Results

I planned to compare my WEKA results with the paper "An Experimental Analysis of Brain Stroke Prediction Using Machine Learning Algorithms" [1], as the model and method used were easy to implement in WEKA and the accuracy they achieved was 95.70% with SMOTE. The problem I encountered was that Random Forest was greyed out when I loaded the data, because my dataset contained string values.

To fix this I did preprocessing. First I removed the string attributes ID and BMI, then I changed the numeric attributes to nominal attributes. The numeric attributes were Age, Hypertension, Heart Disease, Glucose level and Stroke, converted using a filter in the Preprocess tab. From the WEKA 3.8.6 GUI, I opened the tools bar > package manager and downloaded SMOTE. SMOTE was then applied to the Stroke attribute, but it did not improve the accuracy: it was 81% and did not match the accuracy in the research paper.

I checked the paper for the SMOTE percentage, but they did not mention it. I therefore changed the default SMOTE percentage from 100% to 200%, but the accuracy was still only 86%, low compared to the paper. I then changed it from 200% to 300% and the accuracy was 90%, closer, so I increased it to 400%, where the accuracy was 96.60% and matched the paper. This increased the number of STROKE instances from 511 to 3,066, while the number of NO STROKE instances remained the same, as it was not the minority class.

I then ran the classification using the most commonly used model from my research papers, Random Forest, with the default values, and the accuracy was 96.60%. This matched the accuracy of the research paper [1], and all the

other values matched as well. Table II compares my results with the previous studies.

TABLE II. COMPARISON WITH RESULTS FROM PREVIOUS STUDIES

Source	**Evaluation (split / CV)**	**ML algorithm**	**Accuracy**
My results	**10-fold CV**	**RF**	**96.60%**
Singhai et al. [1]	10-fold CV	RF	95.70%
Saputra et al. [3]	10-fold CV	RF	82.07%
My results	**70/30 split**	**RF**	**95.6%**
Shobayo et al. [2]	70/30 split	RF	94.11%

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